

Energy Change in Nuclear Reaction

The energy change in a nuclear reaction is considerably greater than that of a normal chemical reaction. This change can be calculated using Einstein's equation:

$$\Delta E = \Delta mc^2$$

where ΔE is the change in energy, Δm is the change in mass (mass of products - mass of reactants), and c is the speed of light (3.00×10^8 m/s).

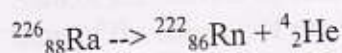
As written, this relation gives the energy change in joules and the mass change in kilograms. Usually small quantities of a sample decay, and the energy change is very large, so it's more common to get an energy change in kilojoules (kJ) corresponding to a mass change in grams. Using the relations

$$1 \text{ kJ} = 10^3 \text{ J and } 1 \text{ kg} = 10^3 \text{ g}$$

Einstein's equation may be rewritten

$$\Delta E \text{ (in kJ)} = 9.00 \times 10^{10} \Delta m \text{ (in grams)}$$

For example, to calculate the ΔE in kJ for the radioactive decay of radium:



when one mole of radium decays, we first calculate Δm for the reaction and then obtain ΔE using the equation.

$$\Delta m = \text{mass of 1 mol } {}^4_2\text{He} + \text{mass of 1 mol } {}^{222}_{86}\text{Rn} - \text{mass of 1 mol } {}^{226}_{88}\text{Ra}$$

$$\Delta m = 4.0015 \text{ g} + 221.9703 \text{ g} - 225.9771 \text{ g}$$

$$\Delta m = -0.0053 \text{ g}$$

Note that Δm may be an extremely small quantity, so it is important to know the masses of products and reactants with a high degree of accuracy in order to know the mass difference to two significant figures.

$$\Delta E \text{ (in kJ)} = 9.00 \times 10^{10} \times (-0.0053)$$

$$\Delta E = -4.8 \times 10^8 \text{ kJ}$$

ΔE in kJ when one gram of radium (one mole weighs 226 g) decays would be:

$$\Delta E = 1 \text{ g Ra} \times (-4.8 \times 10^8 \text{ kJ}) / 226 \text{ g Ra}$$